SECOND APPROACH

Two-detector ensemble for directional people counting

Result: mean counting accuracy rises from 0.6548 to 0.6654 on four camera-separated comparison sequences. The gain is concentrated in the crowded MOT20-01 video; tracking metrics and speed worsen overall.

1. New approach

The baseline uses one YOLO11s person detector fine-tuned for three epochs on MOT17-04/05, followed by our custom ByteTrack and directional line counter. The second approach runs that checkpoint and a COCO-pretrained YOLO11s checkpoint on every frame. It fuses their person boxes before the same tracker and counter. No additional training was needed.

Stage	First approach	Second approach
Detection	Fine-tuned YOLO11s	Fine-tuned + COCO YOLO11s
Box processing	Single-model NMS	Per-model NMS, then box fusion
Tracking/counting	ByteTrack + virtual line	Same ByteTrack + virtual line

2. Motivation

The selected fine-tuned baseline reached only 0.3462 CA on crowded MOT20-01, versus 0.4231 for the COCO-pretrained YOLO11s in the first approach comparison. Fine-tuning helped other camera views. We expected the ensemble to recover additional partially occluded people near the counting line while preserving the fine-tuned detector's strengths.

3. Implementation differences

- Both detectors process the same frame at its configured resolution; person-only confidence floor is 0.05 and per-model NMS IoU is 0.70.
- Boxes from different detectors are grouped when box IoU is at least 0.55. At most one box from each detector enters a group.
- Coordinates are weighted by detector confidence; group confidence is the maximum. Unmatched boxes are retained.
- Fusion runs before the unchanged tracker and counter. The CLI adds --ensemble-with and --fusion-iou; result JSON records both checkpoints.

4. Initial results

Matched GPU reruns on 20 October 2026 used identical scene geometry, tracking, and evaluation settings. Mean values weight each of the four comparison sequences equally. Higher CA, MOTA and IDF1 are better; lower curve MAE is better.

Metric	Baseline	Ensemble	Change
Mean counting accuracy	0.6548	0.6654	+0.0106
Mean curve MAE	1.7021	1.7858	+0.0837
Mean MOTA	0.4113	0.3989	-0.0124
Mean IDF1	0.4324	0.4261	-0.0063
Processing, 2,454 frames	125.4 s	164.7 s	+31% time

Per-sequence counting accuracy

Sequence	Baseline CA	Ensemble CA	Change	Pred IN/OUT	GT IN/OUT
MOT17-02	0.5238	0.5714	+0.0476	10 / 2	16 / 5
MOT20-01	0.3462	0.5385	+0.1923	10 / 6	21 / 5
MOT17-09	0.9231	0.7692	-0.1539	8 / 4	10 / 3
MOT17-11	0.8261	0.7826	-0.0435	7 / 11	11 / 12

The gain is largest on crowded MOT20-01 (+0.1923 CA). CA falls on MOT17-09 (-0.1539) and MOT17-11 (-0.0435). Mean curve MAE rises, MOTA and IDF1 fall, and the ensemble takes about 31% longer. This is a working second approach, not a universal improvement.

Evaluation caveat: the four comparison videos were already used to select the baseline checkpoint in the first submission. These are exploratory post-selection results, not an unbiased final test. The development videos contain detector training frames.

Development sanity check: mean CA was 0.9355 for the single detector and 0.9677 for the ensemble on MOT17-04/05. These figures do not estimate generalization.

Reproduction and source

Run scripts/run_pipeline.py --split test --ensemble-with yolo11s.pt --fusion-iou 0.55 --device 0 --no-render, then scripts/evaluate.py. The complete commands, code, and four machine-readable result files are in the repository:

https://github.com/NikolayTaran/cv_people_counting